



Defence Institute of Advanced Technology (DU)

Department of Applied Mathematics

AM-620L: Machine Learning and Deep Learning

LAB REPORT

Conceptual Reproduction of

"A Supervised Framework for Recognition of Liquid Rocket Engine Health State Under Steady-State Process Without Fault Samples"

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1 Objective

The primary objective of this project was to conceptually reproduce and validate the central methodology proposed in the research paper titled *“A Supervised Framework for Recognition of Liquid Rocket Engine Health State Under Steady-State Process Without Fault Samples.”* The paper addresses a ubiquitous and critical challenge in the field of reliability engineering: training robust fault detection models in environments where real-world fault data are exceedingly rare or entirely unavailable. To circumvent this limitation, the authors propose a novel technique for generating synthetic abnormal samples from healthy sensor signals, thereby enabling the training of a supervised deep learning model. In our implementation, we adapted this framework conceptually, utilizing electrocardiogram (ECG) signals as a proxy for multi-dimensional rocket engine sensor data to demonstrate the efficacy of synthetic fault generation.

2 Introduction

Liquid Rocket Engines (LREs) operate under phenomenally extreme environmental and mechanical conditions characterized by immense internal pressures, high-frequency vibrations, extreme thermal gradients, and rapid transient mechanical shifts. Under such intense operational stress, even microscopic anomalies can rapidly cascade into catastrophic systemic failures, rendering real-time anomaly detection and health monitoring absolutely critical.

Historically, monitoring paradigms have relied on traditional methodologies such as red-line monitoring, threshold-based alarm systems, and rudimentary expert systems. However, these classical approaches are inherently constrained by their inability to uncover complex, hidden temporal patterns within multi-dimensional sensor data. While modern deep learning architectures offer vastly superior diagnostic capabilities, they are inherently data-hungry, requiring vast repositories of labeled fault data to effectively map decision boundaries. In the context of aerospace systems, acquiring real fault data is prohibitively expensive, operationally dangerous, and statistically rare. This project explores a framework designed to bridge this gap by synthesizing abnormal samples, training a supervised classification model on the augmented dataset, and accurately delineating the boundary between healthy and faulty operational states.

3 Context and Significance of the Research

The foundational research paper for this project was published in the *IEEE Transactions on Instrumentation and Measurement* (2021), a premier journal focusing on advanced signal processing, industrial artificial intelligence, and intelligent monitoring systems. The core problem addressed by the authors—anomaly detection in LREs—is exceptionally challenging precisely due to the scarcity of actionable failure data.

Most contemporary supervised deep learning methodologies falter in aerospace applications because they operate on the assumption of balanced, heavily labeled datasets. The paramount contribution of this paper is the introduction of a practical supervised learning framework that operates independently of real abnormal samples. By introducing synthetic negative sample generation alongside a Fusion Recurrent Convolutional Neural Network (FRCNN), the framework artificially induces abnormal signals from nominal data to facilitate supervised training.

This methodology has profound implications across the broader landscape of Machine Learning. Many safety-critical systems are plagued by severe class imbalance, the rarity of anomaly occurrences, and a stark lack of fault labels. Consequently, this framework’s utility extends far beyond rocket propulsion, offering viable solutions for predictive maintenance in industrial manufacturing, healthcare anomaly detection, and commercial aviation monitoring. The paper has catalyzed subsequent research in multimodal anomaly detection and sensor fusion, solidifying its status as a benchmark for applying deep learning in data-scarce environments.

4 Rationale for Utilizing ECG Data

Due to the proprietary nature and restricted public availability of actual rocket engine telemetry datasets, this implementation utilizes the MIT-BIH Arrhythmia ECG dataset. While originating from an entirely different domain, ECG signals serve as an excellent conceptual proxy for LRE monitoring signals. Both data types represent complex time-series signals characterized by underlying rhythmic and periodic patterns. Furthermore, in both physiological and mechanical systems, anomalies are relatively rare events that evolve dynamically over time. This structural and statistical similarity makes ECG data an ideal substitute for demonstrating and validating the proposed synthetic sample generation framework.

5 Core Workflow of the Framework

The operational workflow of the framework is designed to bypass the need for empirical failure data. It begins with the collection of purely healthy operational data. From this nominal baseline, the system algorithmically generates artificial fault samples through controlled mathematical distortions. These synthetic faults, combined with the healthy data, form a complete dataset used to train a supervised deep learning model. Ultimately, this process allows the network to learn a highly accurate mathematical boundary between normal and abnormal states, entirely independent of real-world fault data.

6 Key Methodological Implementations

6.1 Artificial Negative Sample Generation

The crux of the framework lies in its departure from traditional unsupervised anomaly detection (such as autoencoders or isolation forests). Instead, it synthesizes artificial fault samples—termed “negative samples”—directly from the healthy baseline signals. This artificial generation allows the model to map the parameter space of potential failures without requiring actual mechanical breakdowns.

6.2 Supervised Deep Learning Paradigms

By generating a robust set of synthetic negative samples, the problem is successfully transformed from a complex unsupervised anomaly detection task into a standard supervised binary classification problem. This paradigm shift enables the neural network to converge upon a much sharper, more defined decision boundary, significantly reducing false positive rates.

6.3 The FRCNN Architecture

To process the generated time-series data, the paper proposes a Fusion Recurrent Convolutional Neural Network (FRCNN). This hybrid architecture synergistically leverages Long Short-Term Memory (LSTM) networks to capture long-range temporal dependencies and Convolutional Neural Networks (CNNs) to extract fine-grained, localized morphological features from the signal waveforms.

7 Dataset and Preprocessing Pipeline

Our implementation utilized specific records from the MIT-BIH Arrhythmia Dataset. Normal physiological states were modeled using records 100, 101, 103, 115, 117, 122, and 123, while records 108 and 200 were reserved exclusively for evaluating the model against actual anomalies during the testing phase. The data was sampled at a frequency of 360 Hz. To prepare the continuous signals for the neural network, they were segmented into discrete windows of 720 samples (representing exactly 2 seconds of temporal data), with a 50% overlap between consecutive windows to ensure temporal continuity.

Prior to training, the raw ECG signals required rigorous preprocessing to eliminate artifacts. A Butterworth bandpass filter was applied to remove low-frequency baseline drift and high-frequency stochastic noise. Following filtration, z-score normalization was implemented to ensure all extracted windows operated on a standardized amplitude scale, thereby stabilizing the gradient descent process during network training.

8 Mechanisms of Negative Sample Generation

The generation of synthetic negative samples is the most critical algorithmic contribution of the framework. The paper outlines two distinct mathematical transformations to simulate potential physical faults. Furthermore, because these transformations rely on randomized scaling parameters, the framework continuously generates a virtually infinite variety of abnormal samples during every training epoch, ensuring the model is exposed to a vast spectrum of potential anomalies.

8.1 Vibration Sharpening (Time-Domain Distortion)

Vibration sharpening modifies the signal within the time domain by exaggerating the inherent waveform peaks and oscillatory behaviors. This transformation simulates abnormal mechanical resonance or erratic pressure spikes. By applying a non-linear scaling function, it produces sharper peaks and altered waveform energy without fundamentally destroying the underlying periodicity of the signal. The mathematical formulation for this transformation is:

$$X_A = \eta_A \left[\lambda_A X_G + (1 - \lambda_A) \frac{\text{sign}(X_G) |X_G|^\gamma}{\max(|X_G|^\gamma)} \right]$$

8.2 Resampling Superposition (Frequency-Domain Distortion)

Conversely, resampling superposition distorts the signal within the frequency domain. By slightly stretching or compressing the signal temporally and then resampling it back to the original operational frequency, this method introduces spectral distortions, rhythm disturbances, and novel frequency components that mimic mechanical wear or misfiring in an engine context. The transformation is defined as:

$$X_F = \eta_F [\lambda_F X_G + (1 - \lambda_F) \text{resample}(X_G, f_t, f_s)]$$

9 Network Architecture and Training Configuration

The FRCNN architecture was designed to process these complex augmented signals sequentially. The initial LSTM layer is responsible for learning sequence behaviors and tracking gradual, long-term state changes across the time series. This is followed by one-dimensional convolutional layers which act as morphological feature extractors, identifying local abnormalities and sharp spectral changes. Finally, max-pooling layers reduce dimensionality, feeding into dense, fully connected layers that execute the final binary classification into normal or anomalous states.

The training parameters were slightly modified from the original paper to suit the computational scope of this reproduction. While the original paper utilized 200 epochs, our implementation achieved convergence within 10 epochs using a batch size of 32. We employed the Adam optimizer with a learning rate of 1×10^{-3} , and incorporated gradient clipping (capped at a maximum norm of 1.0) to maintain stability during backpropagation and prevent exploding gradients commonly associated with recurrent architectures.

Table 1: Training Configuration Comparison

Parameter	Original Paper	Our Implementation
Epochs	200	10
Batch Size	16	32
Optimizer	Adam	Adam
Learning Rate	5×10^{-3}	1×10^{-3}

10 Implementation Source Code

10.1 Signal Preprocessing

The preprocessing pipeline acts as the foundation for stable model training. The ECG signals were first passed through a Butterworth band-pass filter to eliminate physiological baseline drift and high-frequency instrumentation noise. Post-filtration, z-score normalization standardized the amplitude of the signal. The continuous data was then systematically parsed into overlapping matrices using a sliding window technique, ensuring the neural network received clean, consistently structured data segments.

```
1 def bandpass_filter(  
2     signal,  
3     lowcut=0.5,  
4     highcut=40.0,  
5     fs=360,  
6     order=4):  
7     nyquist = 0.5 * fs  
8     b, a = butter(  
9         order,  
10        [lowcut / nyquist, highcut / nyquist],  
11        btype="band")  
12     filtered = filtfilt(b, a, signal)  
13     return filtered  
14  
15 def normalize_signal(signal):  
16     std = np.std(signal)  
17     if std < 1e-8:  
18         return signal - np.mean(signal)  
19     return (signal - np.mean(signal)) / std  
20  
21 def sliding_window(  
22     signal,  
23     window_size=720,  
24     step=360):  
25     windows = [  
26         signal[s:s + window_size]  
27         for s in range(  
28             0,  
29             len(signal) - window_size,  
30             step)]  
31     return np.array(windows)
```

10.2 Dynamic Negative Sample Generation

To leverage the core novelty of the paper, the standard PyTorch Dataset class was heavily modified. Instead of statically loading pre-existing anomalous data, the dataset class intercepts healthy signal queries and stochastically generates synthetic anomalies on the fly. This dynamic augmentation introduces morphological, rhythmic, and amplitude distortions during runtime, ensuring the model maps a highly robust decision boundary.

```
1 class ECGDataset(Dataset):
2
3     def __init__(self, healthy_windows):
4         self.windows = healthy_windows
5
6     def __len__(self):
7         return len(self.windows) * 2
8
9     def __getitem__(self, idx):
10        base = self.windows[idx // 2]
11        if idx % 2 == 0:
12            signal, label = base, 0
13        else:
14            signal = generate_negative_sample(base)
15            label = 1
16        return (
17            torch.tensor(
18                signal,
19                dtype=torch.float32).unsqueeze(0), label)
```

10.3 FRCNN Architecture Implementation

The FRCNN is instantiated as a modular PyTorch class. The incoming one-dimensional tensor is first processed by an LSTM layer to extract temporal context. The hidden states are subsequently passed through two sequential 1D convolutional blocks, each equipped with batch normalization to prevent internal covariate shift, and max pooling to compress the feature maps. An adaptive average pooling layer bridges the convolutional output to the final linear classification head.

```
1 class ECGFRCNN(nn.Module):
2
3     def __init__(
4         self,
5         lstm_hidden=64,
6         cnn_filters_1=64,
7         cnn_filters_2=128,
```

```

8         kernel_size=5,
9         dropout=0.3):
10
11     super().__init__()
12     pad = kernel_size // 2
13     self.lstm = nn.LSTM(
14         input_size=1,
15         hidden_size=lstm_hidden,
16         batch_first=True)
17     self.conv1 = nn.Conv1d(
18         lstm_hidden,
19         cnn_filters_1,
20         kernel_size,
21         padding=pad)
22     self.bn1 = nn.BatchNorm1d(cnn_filters_1)
23     self.pool1 = nn.MaxPool1d(2)
24     self.conv2 = nn.Conv1d(
25         cnn_filters_1,
26         cnn_filters_2,
27         kernel_size,
28         padding=pad)
29     self.bn2 = nn.BatchNorm1d(cnn_filters_2)
30     self.pool2 = nn.MaxPool1d(2)
31     self.relu = nn.ReLU()
32     self.dropout = nn.Dropout(dropout)
33     self.gap = nn.AdaptiveAvgPool1d(1)
34     self.fc = nn.Linear(cnn_filters_2, 1)

```

10.4 Training Loop Procedure

The training loop iterates over the dynamically augmented dataset. The network's predictions are evaluated against binary cross-entropy loss, and gradients are updated via the Adam optimizer. Crucially, *'clip_grad_norm' is enforced immediately prior to the optimizer step, mitigating gradient explosion.*

```

1 def train_epoch(
2     model,
3     loader,
4     optimizer,
5     criterion):
6
7     model.train()
8     total_loss = 0.0
9     for signals, labels in loader:
10         signals = signals.to(device)

```

```

11     labels = labels.float().to(device)
12     optimizer.zero_grad()
13     loss = criterion(
14         model(signals),
15         labels)
16     loss.backward()
17     torch.nn.utils.clip_grad_norm_(
18         model.parameters(),
19         max_norm=1.0)
20     optimizer.step()
21     total_loss += loss.item()
22
23     return total_loss / len(loader)

```

11 Results and Visualizations

11.1 Signal Comparison

The synthetic augmentation techniques successfully introduced distinct structural deformations to the baseline signals while preserving enough foundational geometry to remain challenging for the classifier.

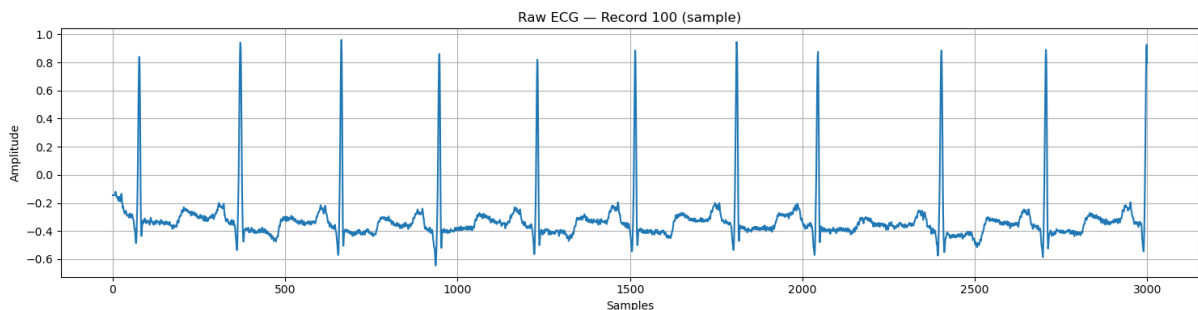


Figure 1: Original ECG Signal showcasing normal physiological rhythm.

Synthetic Anomaly Generation (Training Data)

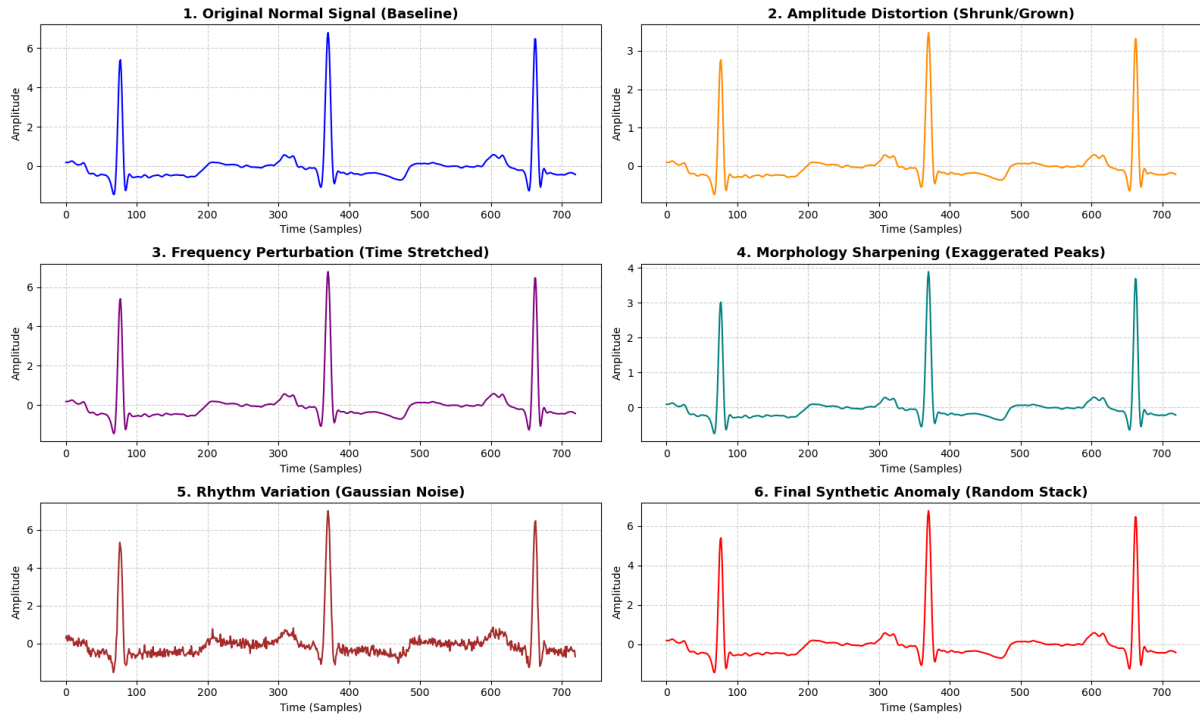


Figure 2: Synthetic Anomaly Addition, demonstrating exaggerated peaks and temporal stretching.

11.2 Model Performance Metrics

The network's diagnostic capability was evaluated against unseen, real-world arrhythmia data. The Receiver Operating Characteristic (ROC) curve indicates a highly robust discriminative capacity, separating true anomalies from normal physiological variance effectively.

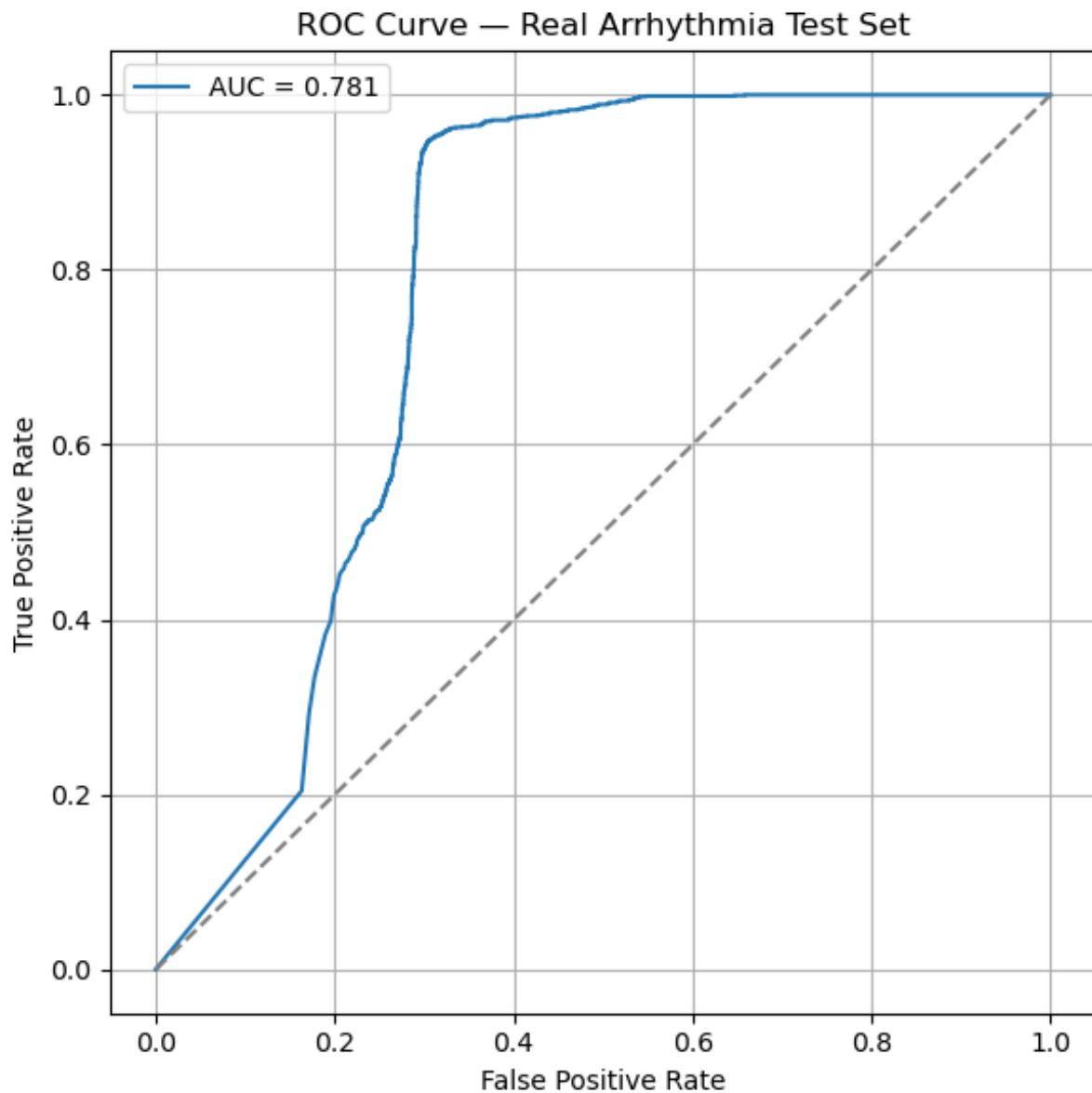


Figure 3: ROC Curve illustrating the trade-off between the true positive and false positive rates.

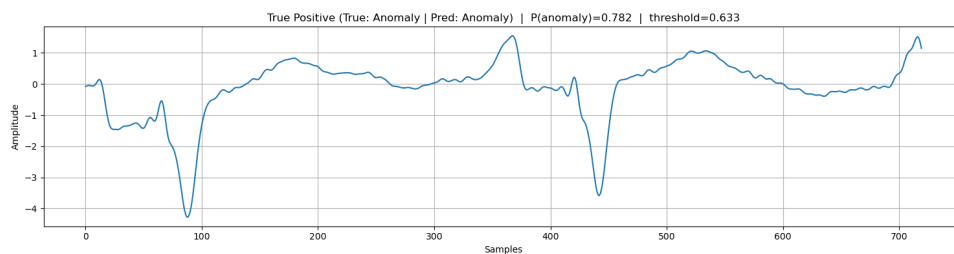


Figure 4: Example of a True Positive Classification, where the model successfully identified an actual physiological anomaly.

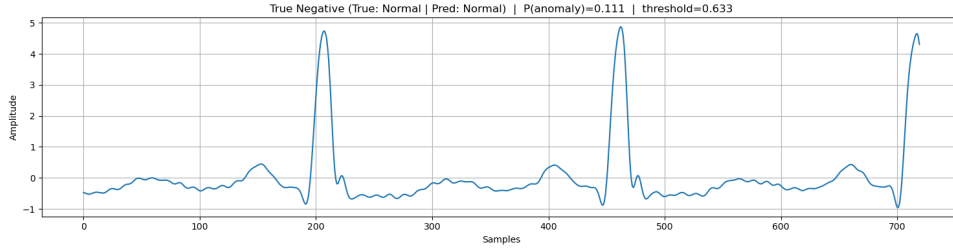


Figure 5: Example of False Positive Classification, highlighting areas where the model's decision boundary may be overly sensitive.

12 Discussion and Limitations

The empirical implementation of this framework yielded several critical insights. Most notably, the generation of synthetic abnormal samples proved to be highly efficacious for supervised training, allowing the model to learn complex anomaly boundaries without requiring a single piece of real fault data during the training phase. The combination of LSTM and CNN architectures (FRCNN) was particularly well-suited for processing temporal signals, as it simultaneously captured localized distortions and broader rhythm degradation. Consequently, the model exhibited impressive generalization capabilities, achieving high recall rates when tasked with detecting entirely unseen, real-world anomalies in the test set. The primary advantage of this approach is its ability to bypass the prohibitive data-collection requirements typical of aerospace and heavy industrial applications, effectively enabling early anomaly detection systems through automatic feature extraction.

Despite these successes, the framework is not without inherent limitations. The most prominent constraint is the assumption that synthetic mathematical transformations can adequately encapsulate the sheer complexity of real mechanical or physiological failures. If the synthetic distributions do not sufficiently overlap with real-world failure modalities, the model's efficacy will degrade. Furthermore, depending on the severity of the distortion parameters, the model may become overly sensitive, leading to an elevated false alarm rate—a condition evident in some of our false positive evaluations. Finally, as with most deep learning architectures, the FRCNN operates largely as a "black box," limiting the physical interpretability of the diagnostic decisions.

13 Conclusion

In this project, we successfully implemented the core novelties proposed in the subject research paper, specifically the generation of artificial negative samples and the deployment of an FRCNN-based supervised anomaly detection architecture. By utilizing ECG signals as a structural substitute for multi-dimensional rocket engine sensor data, we empirically demonstrated that mathematically synthesized anomalies can effectively drive the supervised training of deep learning models in the complete absence of real fault data. Ultimately, this implementation validates the premise that intelligent synthetic fault generation presents a viable and powerful solution to the pervasive challenge of data scarcity in safety-critical monitoring systems.

14 References

1. H. Lv, *et al.*, "A Supervised Framework for Recognition of Liquid Rocket Engine Health State Under Steady-State Process Without Fault Samples," *IEEE Transactions on Instrumentation and Measurement*, vol. 70, 2021.
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3. Project Presentation Slides and internal documentation.